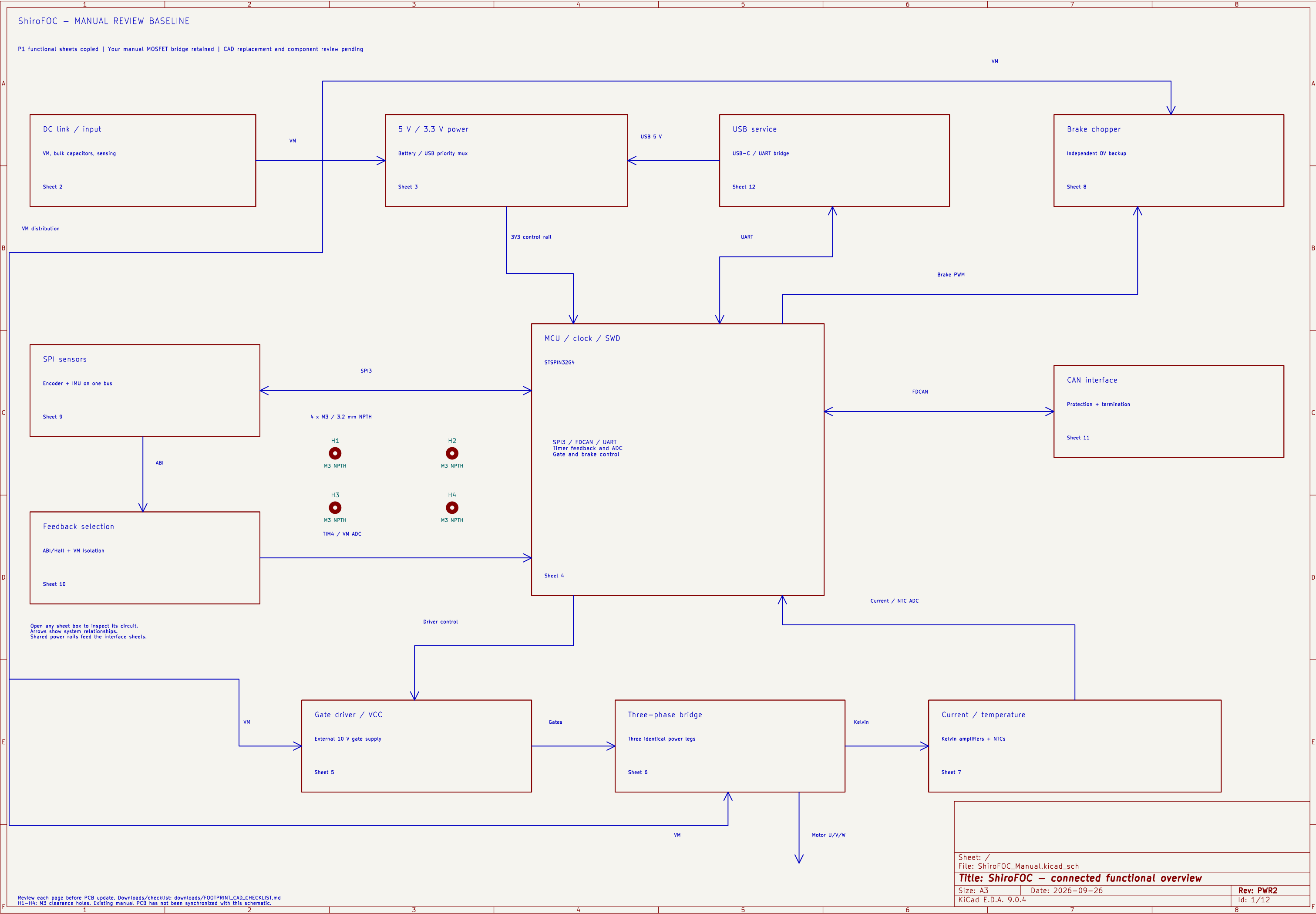
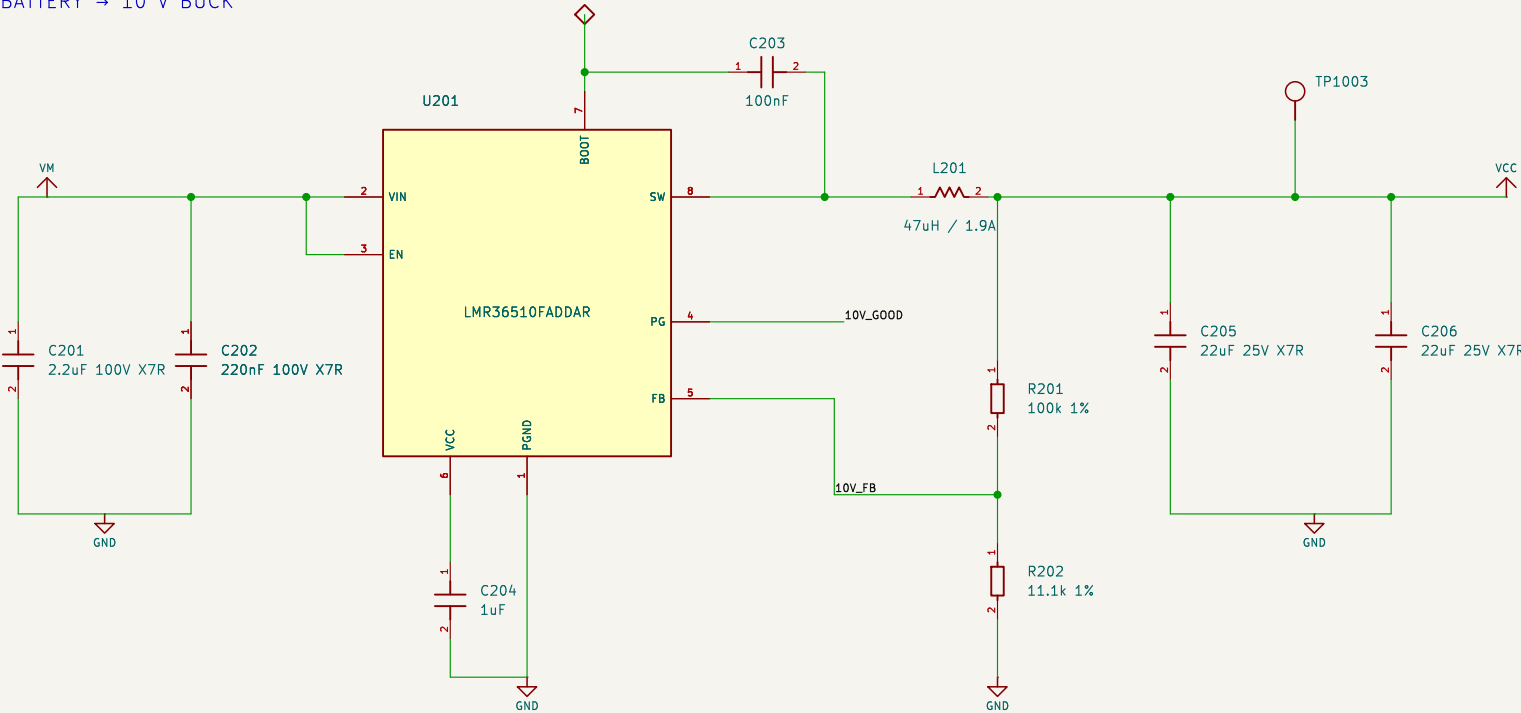




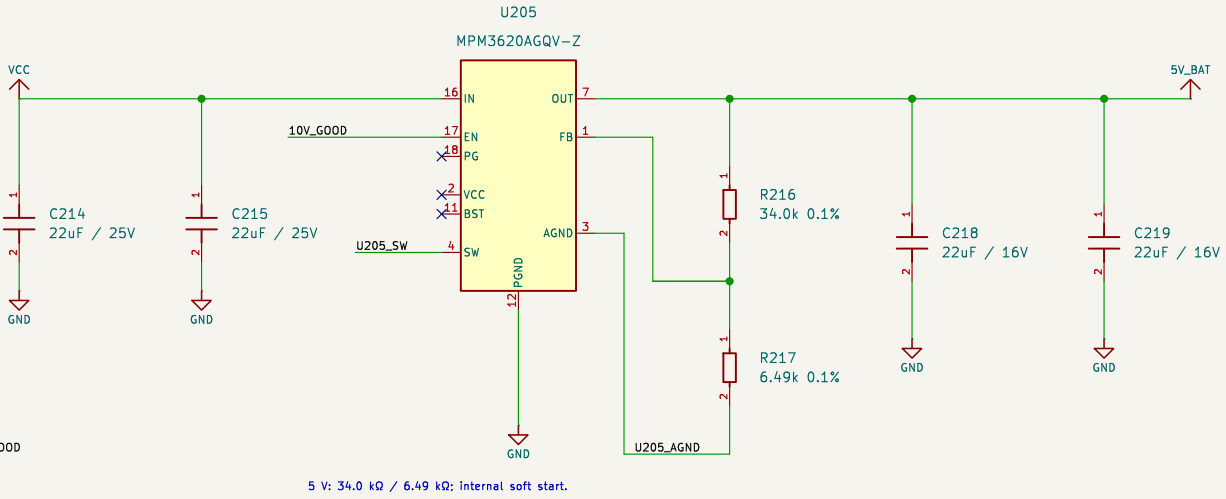
AUXILIARY POWER – USB HAS PRIORITY

BATTERY → 10 V BUCK



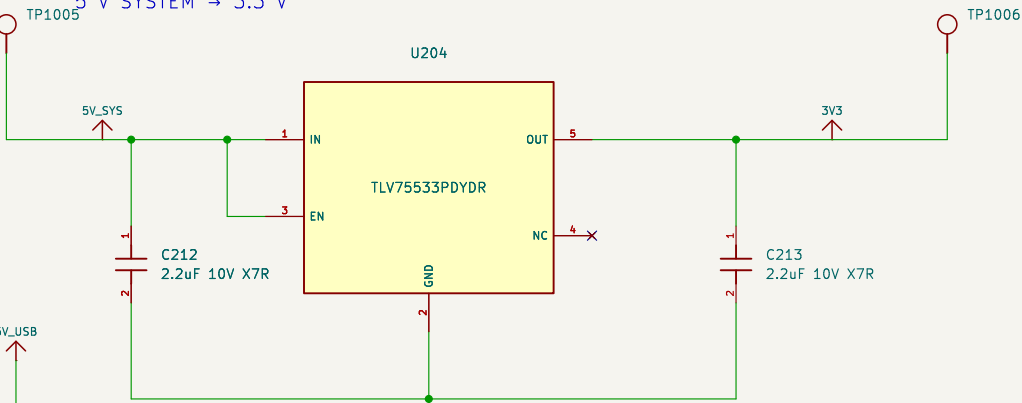
10 V → 5 V / INTEGRATED INDUCTOR

U205 VCC is its internal 4.9 V output; leave open.
AGND serves only the FB divider; joined to PGND inside U205.



5 V: 34.0 kΩ / 6.49 kΩ; internal soft start.

5 V SYSTEM → 3.3 V

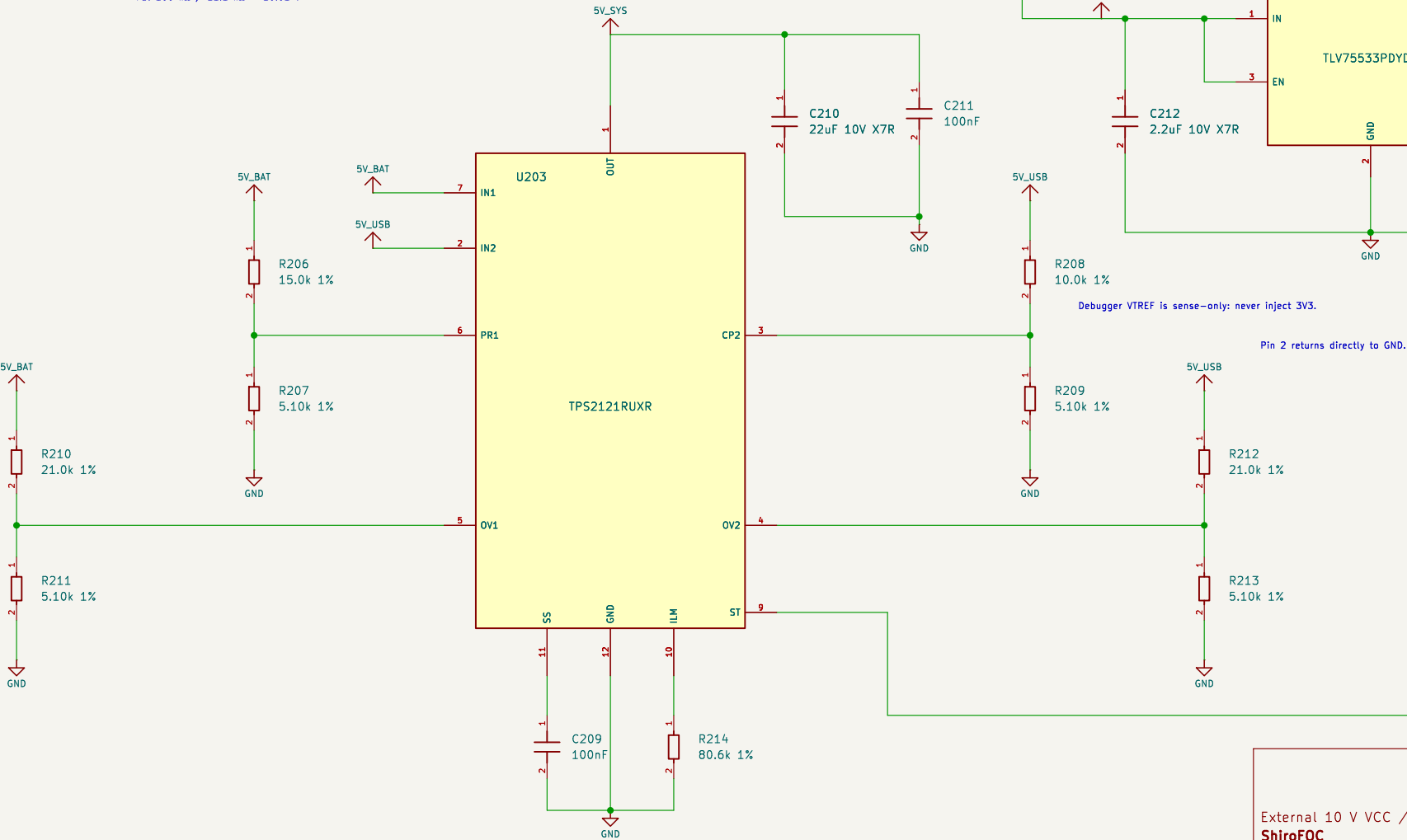


Debugger VTREF is sense-only; never inject 3V3.

Pin 2 returns directly to GND.

USB-PRIORITY POWER MUX

5V_USB arrives from the USB service sheet.



At 5 V: PR1 = 1.269 V; CP2 = 1.689 V.

USB wins the priority comparison. OV ≈ 5.43 V.

80.6 kΩ sets approximately 1.5 A current limit.

VCC is the external 10 V gate supply. Local caps remain at the driver.

U201 PG enables U205 through 100 kΩ / 100 kΩ; EN stays below 6 V.

External 10 V VCC / integrated-inductor 5 V supply

ShiroFOC

Sheet: /5 V / 3.3 V power/

File: 02_Auxiliary_Power.kicad_sch

Title: 10 V / 5 V / 3.3 V auxiliary supplies

Size: A3

Date: 2026-09-26

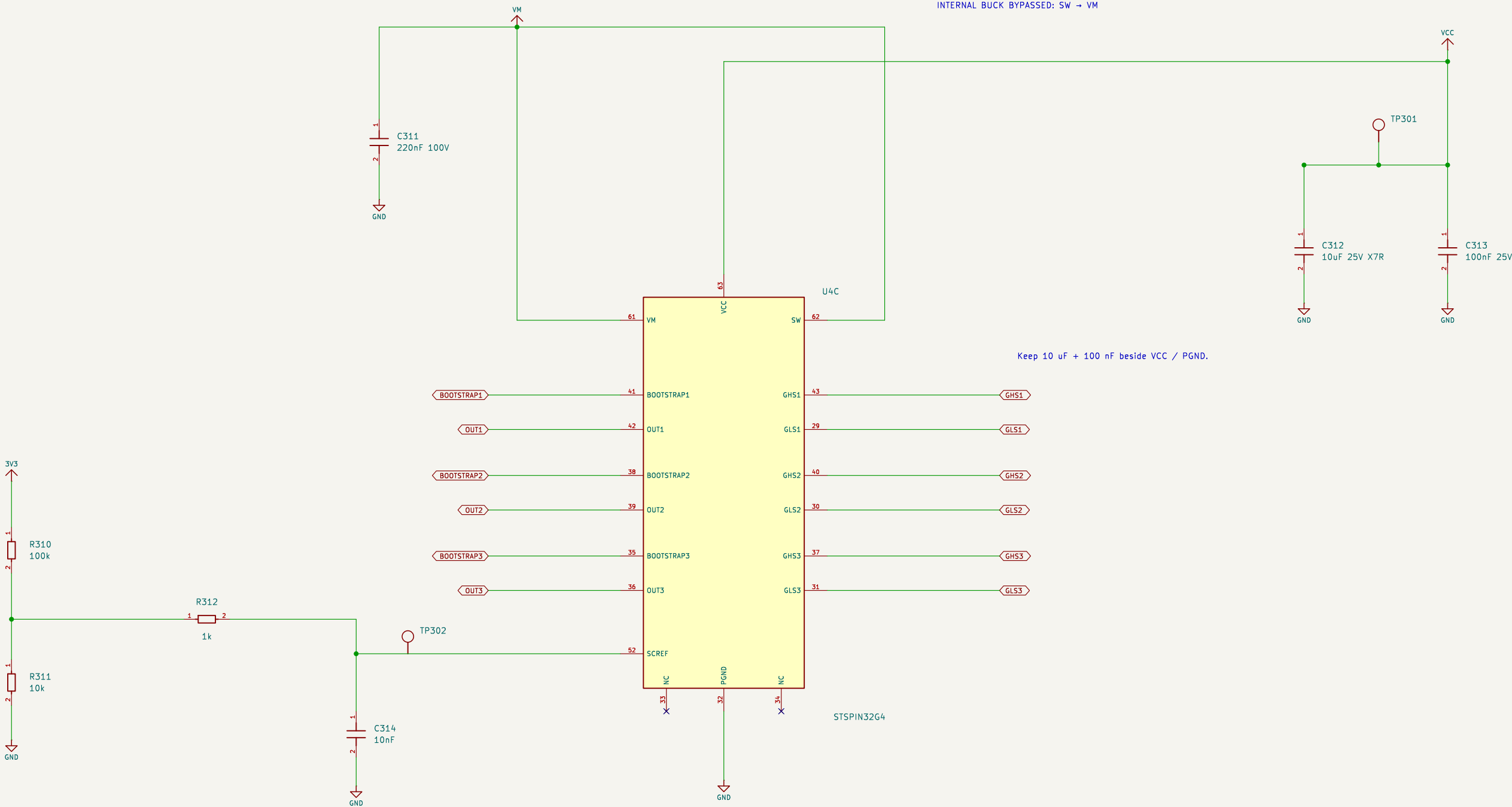
Rev: PWR3

KiCad E.D.A. 9.0.4

Id: 3/12

GATE DRIVER / EXTERNAL 10 V SUPPLY

VDS FAULT REFERENCE



SCREF is a VDS fault threshold (~0.30 V), not a precision phase–current limit. Configure COMP1/2/4 and TIM1 break before enabling PWM.

External VCC = 10 V from the auxiliary sheet. SW tied to VM per ST Figure 7. REGIN and REG3V3/VDD use external 3V3 (unit B). USB alone does not power the gates.

External 10 V VCC / integrated–inductor 5 V supply

ShiroFOC

Sheet: /Gate driver / VCC/

File: 04_Gate_Driver.kicad_sch

Title: STSPIN gate driver / external 10 V

Size: A3

Date: 2026–09–26

Rev: PWR2

KiCad E.D.A. 9.0.4

Id: 5/12